

ENEL301 – Fundamentals of Engineering
Economics and Management

Sustainable Development

&

Systems Thinking



Workshop Goal & Instructions

Goal: get to know your team for the sustainability assignment, including their different perspectives and to develop a basic causal-loop diagram

Instructions in brief:

1. **Find your team** – Find your other group members & greet!
2. **Images of Development** – Rank in order from most to least developed, prepare to discuss
3. **Causal Loop Diagram** – develop a CLD for working while you study

Images of Development

- Rank these from “most developed” to “least developed”



Lockheed Martin F-35 Lightning II



International Space Station



Pyramid complex, Giza, Egypt



Combine-harvester harvesting a crop



Mount Aspiring National Park, NZ



Holi; the Hindu Festival of Colours, Love and Spring

- Be ready to discuss how you arrived at your ranking
- It's OK to politely disagree!

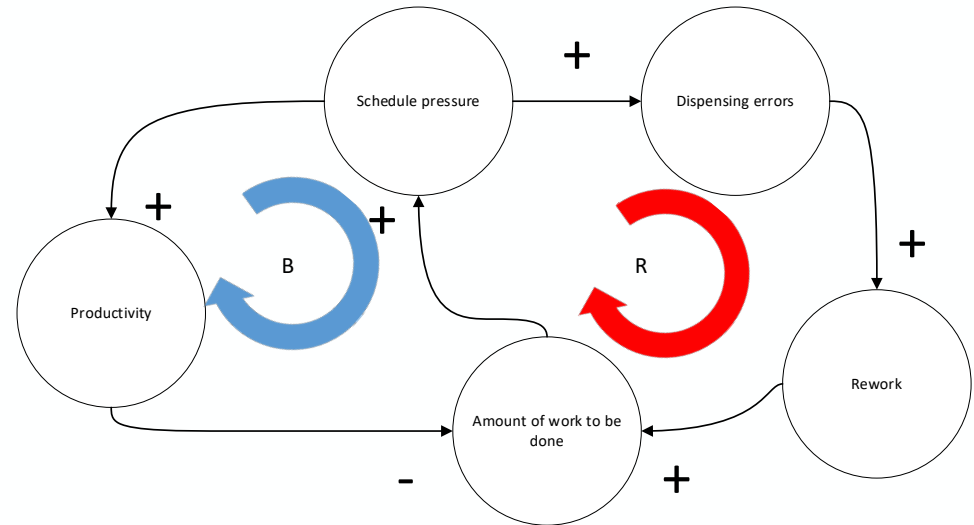
Images of Development

Take-away messages

- Definition of development depends on individual and collective perspectives of (e.g. Technological? Societal? Environmental? Economic?)
- What is 'developed' will vary with context
 - “Pictures don't tell the full story”
 - Local vs global
 - Technology is ever advancing – what we consider as 'developed' changes
 - Protection can be a form of development
- Balances between social, environmental, economic impacts and benefits are challenging, and sometimes progress is very slow or we regress

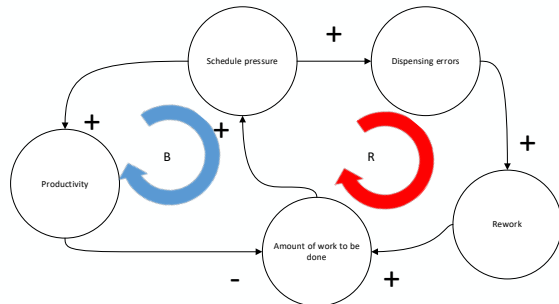
Causal loop diagrams

- Create variable names → nouns
- Draw links
- Identify causal effects
 - S = Same OR +
 - O = Opposite OR -
- Identify and label any loops
 - B = Balancing
 - R = Reinforcing
- Use loopy: <https://ncase.me/loopy/v1.1/>



Create a Causal Loop Diagram for “commuting to university by bicycle”

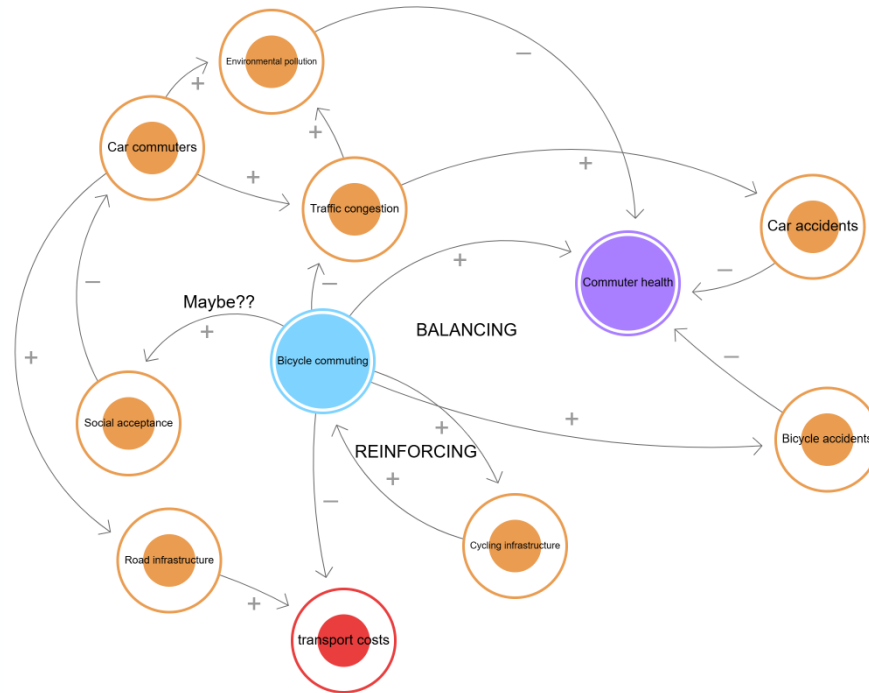
- Create variable names → nouns
- Draw links & Identify causal effects
 - S = Same OR +
 - O = Opposite OR -
- Identify and label any loops
 - B = Balancing
 - R = Reinforcing



- Core nodes:
 - Bicycle commuting
 - Health of commuter
 - Transport costs
 - Car commuters
- Include other nodes as needed, including your own socio-economic outcomes
- Link the nodes together in a CLD, including identify direction of causal effects, and a balancing (B) and reinforcing loop (R)
- Draw manually or use loopy:
<https://ncase.me/loopy/v1.1/>
- **Link three outcomes to a relevant UN Sustainable Development Goal, Target, and Indicator**

An example answer

Causal Loop Diagram



Limitation:

- Descriptive / qualitative; does not account for strength of causal effects

Tip for your assignment:

- Describe and justify the relationships and loops

An example answer

Relevant UN SD Goal, Target and Indicator

Commuter health

SDG 3: Good Health and Well-being: Ensure healthy lives and promote well-being for all at all ages.

Target 3.4: By 2030, reduce by one third premature mortality from non-communicable diseases through prevention and treatment and promote mental health and well-being.

Indicator 3.4.1: Mortality rate attributed to cardiovascular disease, cancer, diabetes or chronic respiratory disease

Causal Loop Diagrams

Take-away messages

- A useful systems thinking tool to identify
 - Potential outcomes, including the unintended outcomes
 - Directional relationships occurring within your system
- Inputs, processes or outputs within your system sometimes be reinforced or balanced
- Qualitative – does not quantify or account for the strength of the different relationships
- You will be expected to apply this in your assignment